

Zhiyuan Li

Postdoctoral Researcher

Aalto Robot Learning
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Research Interests

Multi-Agent Systems, Reinforcement Learning, Robotics, Foundation Models

Education and Work Experience

Aalto Robot Learning, Aalto University <i>Postdoctoral Researcher</i>	Aug 2024 – Present <i>Espoo, Finland</i>
University of Electronic Science and Technology of China <i>Ph.D. in Computer Science and Technology</i>	Sep 2019 – July 2024 <i>Chengdu China</i>
North China Electric Power University <i>Bachelor in Computer Science and Technology</i>	Sep 2015 – July 2019 <i>Beijing China</i>

Publications

Why Cross-Skeleton Retargeting Is Non-Identifiable: Structural Limits of Generative Motion Models

Zhiyuan Li, Wenyan Yang, Pekka Marttinen, and Joni Pajarinen.
[NeurIPS'26] *The Fortieth Annual Conference on Neural Information Processing Systems, 2026.*

Sparsely Supervised Diffusion

Wenshuai Zhao, Zhiyuan Li, Yi Zhao, Mohammad Hassan Vali, Martin Trapp, Joni Pajarinen, Juho Kannala, and Arno Solin.
[NeurIPS'26] *The Fortieth Annual Conference on Neural Information Processing Systems, 2026.*

Rethinking Temporal Consistency in Video Object-Centric Learning: From Prediction to Correspondence

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Pekka Marttinen, and Joni Pajarinen.
[ICML'26] *The Forty-Third International Conference on Machine Learning, 2026.*

Learning Progress Driven Multi-Agent Curriculum

Wenshuai Zhao, Zhiyuan Li, and Joni Pajarinen.
[ICML'25] *The Forty-Second International Conference on Machine Learning, 2025.*

AgentMixer: Multi-Agent Correlated Policy Factorization

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.
[AAAI'25] *Thirty-Ninth AAAI Conference on Artificial Intelligence, 2025.*

Optimistic Multi-Agent Policy Gradient for Cooperative Tasks

Wenshuai Zhao, Yi Zhao, Zhiyuan Li, Juho Kannala, and Joni Pajarinen.
[ICML'24] *The Forty-first International Conference on Machine Learning, 2024.*

Coordination as Inference in Multi-Agent Reinforcement Learning

Zhiyuan Li, Lijun Wu, Kaile Su, Wei Wu, Yulin Jing, Tong Wu, Weiwei Duan, Xiaofeng Yue, Xiyi Tong, and Yizhou Han.
[Neural Networks'24] *Neural Networks, 2024.*

Backpropagation Through Agents

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.
[AAAI'24] *Thirty-Eighth AAAI Conference on Artificial Intelligence, 2024.*

Online Coordinated NFV Resource Allocation via Novel Machine Learning Techniques

Zhiyuan Li, Lijun Wu, Xiangyun Zeng, Xiaofeng Yue, Yulin Jing, Wei Wu, and Kaile Su.

[TNSM'22] *IEEE Transactions on Network and Service Management*, 2022.

Closed-Loop Vision-Language Planning for Multi-Agent Coordination

Zhiyuan Li, Wenshuai Zhao, and Joni Pajarinen.

[SE@AAMAS'26] *The 25th International Conference on Autonomous Agents and Multiagent Systems*, 2026.

Adaptive graph attention networks with interactive learning for attributed graph clustering

Weiwei Duan, Luping Ji, Lijun Wu, Qi Deng, and Zhiyuan Li

[EAAI'26] *Engineering Applications of Artificial Intelligence*, 2025

Multi-agent neighborhood coordinated and holistic optimized actor-critic framework for adaptive traffic signal control

Qi Deng, Lijun Wu, Zhiyuan Li, Kaile Su, Wei Wu, and Weiwei Duan.

[Applied Intelligence'25] *The International Journal of Research on Intelligent Systems for Real Life Complex Problems*, 2025

Bridging the Embodiment Gap: Disentangled Cross-Embodiment Video Editing

Zhiyuan Li, Wenyan Yang, Wenshuai Zhao, Yue Ma, Yuanpeng Tu, Pekka Marttinen, and Joni Pajarinen.

arXiv preprint, arXiv:2605.03637, 2026.

Do We Really Need the Default Recipe for Video Object-Centric Learning?

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Arno Solin, Juho Kannala, Pekka Marttinen, and Joni Pajarinen.

[TPAMI] *IEEE Transactions on Pattern Analysis and Machine Intelligence*. (Under Review)

Teaching Experience

Main teaching assistant for ELEC-E8125 Reinforcement Learning, Department of Electrical Engineering and Automation, Aalto University, Finland (243 participants); coordinated the 12-person TA team, exercises, student project support, technical Q&A, and grading.

Research Funding

Scholarship under China Scholarship Council, Ministry of Education of the P. R. China, EUR 16,200, 2022.

Participated in grant application preparation for the 2026 International Collaboration in High-Performance Computing call of the Research Council of Finland, with Principal Investigators Joni Pajarinen (Aalto University) and Ville Hautamaki (University of Eastern Finland).

Awards & Honors

Academic Scholarship	2019-2022
<i>University of Electronic Science and Technology of China</i>	
Honorable Mention of Mathematical Contest in Modeling	2017

Community Activity

Area Chair for ICLR 2027.

Conference reviewer for AAAI, ICML, NeurIPS, ICLR, ECCV, and AAMAS.

Supervision

Mentored and collaborated closely with Ph.D. researchers Wenshuai Zhao (graduated 2025) and Yi Zhao (expected graduation 2026).